

Raptor Geodatabase

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Introduction

Raptors are birds of prey that occupy the top of the food chain. They are key species in ecosystems, and their well-being often reflects the health of ecosystems they occupy. As an apex predator, raptors are incredibly vulnerable to environmental change. Raptors are relatively easy to track in comparison to smaller terrestrial organisms, as these larger birds can be spotted, banded, and studied as they fly above research sites. Every year during the fall and spring, millions of birds migrate south along mountain ridges and river valleys, creating a predictable, highly visible opportunity for cyclical data collection. The Hawk Migration Association of North America maintains a database of ‘hawkwatches’, wherein observers watch migratory birds and upload the data they track online. There are two HawkWatches in the state of Iowa: Hitchcock Hawkwatch located outside Omaha, viewing the Missouri River, and Grammar Grove Hawkwatch in Marshalltown, viewing the Iowa River (which eventually funnels into the Mississippi River Valley). Each HawkWatch has 38 years of data, but for this project we selected twenty years' worth (2004-2024) due to recency and ease of viewership. This project sought to understand the temporal migratory patterns of raptors in Iowa, contrasting the quantity and distribution of species in the Missouri and Mississippi River Valleys.

A plethora of raptor species are included in the Hawkwatch collection protocol. Every species is currently federally recognized as being “Least Concern”; however, not every species has had a stable history with some species needing significant interventions. Iowa maintains a list of animals that are of conservation interest. Of this group, four raptors are recognized: the Red Shouldered hawk and Northern Harrier hawk are endangered, and the Peregrine falcon and Bald eagle are special concerns. The unique position of these species raises their importance in this project.

Fundamentally, we want to analyze data from the past to build a foundation for understanding how species may be vulnerable in the future. We have focused on a few main questions to explore: How has the rate of migration for various species of raptors along the Missouri River compared to the Iowa River, and how has this changed over time? What are raptor species preferred conditions: landcover, weather, and proximity to bodies of water? How does that correlate with migration patterns?

We selected data from the typical migratory months of September, October, November, and December for both sites. By monitoring the counts in these months over a twenty-year period, we can identify patterns between specific species counts and other conditions. Similarly, the database can also be used to identify if species migration counts have increased; while this cannot directly say that total bird populations have increased, it can be said that conservation efforts in Iowa are at least not harmful. This database is intended for keeping track of hawk sightings and viewing patterns in their migration based on observed sightings. It is possible for this data to have flaws related to efficacy in comparison; if there are more observers at a given site, it is possible to count more birds or extend hours of operation. Visual identification is also never perfect, and some species may be easily confused and over- or undercounted. The intended audience would be a casual viewer who is interested in general bird migration. The database is

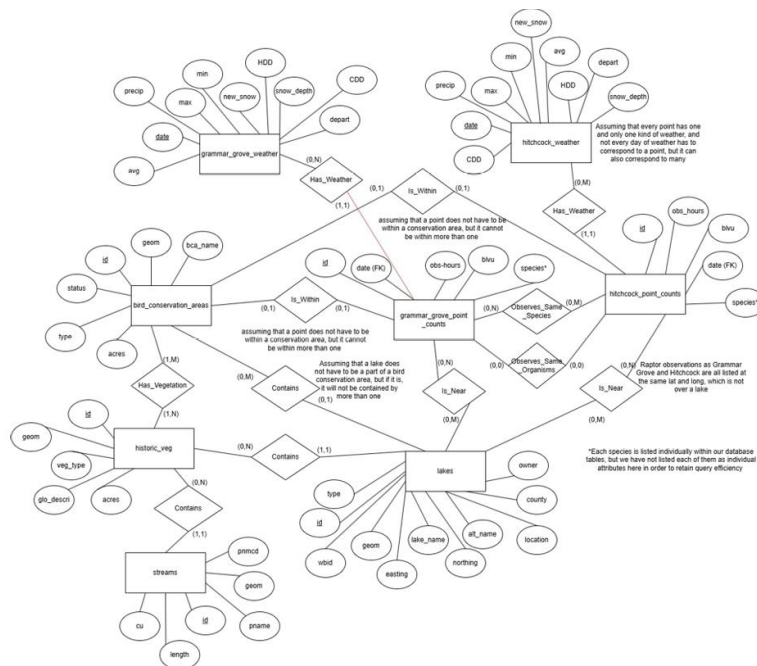
restricted to those who can edit and update the database because hawk identification can be difficult, and false identifications would lead to inaccuracies.

Database Design and Implementation

To analyze raptor sighting data from Hawkwatch and understand migratory patterns, we had to confine the data we were looking at and include all factors. We downloaded data from two locations in Iowa from 2004 to 2024, Hitchcock Hawkwatch and Grammar Grove Hawkwatch. By confining our data to Iowa for a select number of years, we were able to analyze our data better and maintain query efficiency. Other factors we included that would correlate with the number of sightings in each location were historic vegetation, bird conservation areas, locations of major water sources, and weather data such as temperature and precipitation.

Our spatial data was downloaded from the Iowa Geospatial Clearinghouse. All spatial data was changed into EPSG:26915 – NAD83 / UTM zone 15N in QGIS to make sure all data was spatially comparable. The Historic Vegetation dataset contains landcover classes collected from 1832 – 1859, with classes like timber, prairie, field, swamp, pond, and meadow. Each of the 15,500 rows has an ID, a geometry, acreage, vegetation type, and a description.

Weather data was collected from the online tabular database maintained by weather.gov and NOAA. Grammar Grove's weather is represented by weather data collected in Marshalltown, IA, while Hitchcock's weather is represented by data collected in the Omaha Epply Airfield (which was the closest survey location). The weather data contains (in Degrees Fahrenheit) maximum temperature, average temperature, departure from normal, heating degree days, cooling degree days, precipitation, new snow, and snow depth. These were all imported into our database as X (integer, float, go back and look). This data was downloaded for the months of September, October, November, and December from 2004 – 2024.



The Entity Relationship Diagram we created features all eight of our datasets (or entities) with their own attributes. The diagram connects the entities by relationship. For most of the

entities, the primary key is the unique id present in each table. The weather data used the date as the primary key. The date also serves as a foreign key in the Grammar Grove and Hitchcock point count entities. Although there are many relationships present in our database, the clearest relationships were between the Hitchcock Hawk Watch count, the Grammar Grove Hawk Counts, and their respective weather data. Every Hawk Watch raptor observation must be associated with one and only date and set of weather. The same date and weather, however, can be associated with zero or many raptor observations. Most other relationships within our database are spatial, such as whether an area 'X' is within or near an area 'Y'. As an example, Grammar Grove raptor observations have the potential to be within one conservation area, but one observation cannot be counted as within more than one conservation area. Raptor observations can also be found within zero conservation areas, and we will see that is the case for the observations at Hitchcock and Grammar Grove.

Analysis

The first step in data collection was to download the Hawkwatch data. An excel spreadsheet was created for both the Grammar Grove and Hitchcock sites. Online records available on Hawkcount.org were manually copied month-by-month over the twenty-year period into those spreadsheets. Empty rows were selected and converted into zeroes. The same methodology was used to collect weather data from the NOWData – NOAA Online Weather Data portal. A location closest to the count site, and daily data for a month, were selected. These items were then copied into respective spreadsheets and adjusted for the inclusion of zeroes in blank rows.

Tables were created for each site for counts and weather in PGAdmin with correct columns, data types, permissions, restrictions, primary keys, and null limitations. Spatial datasets were imported into QGIS, where they were projected into EPSG:26915 if not already in the correct datum. These were then brought into PGAdmin.

Four tables were made to achieve the first level of normalization. In the main count tables, all data was listed in columns while dates represented rows. This was unideal. A table for baea, haha, pefa, and rsha were created. Each table has a date, which is the primary key and not null, of the data-type 'date'. They also had gg_obs and h_obs, for the observed hours (double), and gg_count and h_count for the counts. This was repeated for each bird, and these were set as integers. A query was written to transfer the data to the new table from the existing table.

Query Query History

```
1  INSERT INTO project.baea (date, g_obs, g_count)
2      SELECT date, obs, baea
3      FROM project.gg_count
4
```

After the data from the first site was imported, the table then had to be updated to prevent an error where the date column was registered as either duplicate or null:

```

1 UPDATE project.baea AS b
2 SET
3     h_obs = h.obs,
4     h_count = h.baea
5 FROM project.hitchcock_count AS h
6 WHERE b.date = h.date;
7

```

Now, there are four normalized tables to work with.

In addition to data normalization, we also used SQL code to create a wide variety of queries that help us answer our research questions. Spatially, we created queries that returned the distance, in kilometers, of each bird conservation area in Iowa to each Hawk Watch site. To return distance in meters and not degrees, we ensured that all spatial datasets were projected into our chosen coordinate reference system of EPSG:26915 – NAD83 / UTM zone 15N. Although units would have been accurately returned in meters, we then divided the distance values by 1000 to return kilometers, which is much easier to interpret on a larger scale.

Next, we turned our focus to the weather data to see if there were any relationships between the raptor species and the weather on the day they were observed. Instead of focusing on all twenty-two raptors observed at Hitchcock and Grammar Grove, we chose to only pull data for the endangered Red Shouldered hawk and Northern Harrier hawk, and the Peregrine falcon and Bald eagle of special concern. In the following queries, we return the rate of these raptor sightings at Hitchcock Hawk Watch and Grammar Grove Hawk Watch, in the years 2004 and 2024 (beginning and end years of our current database), when temperatures are below seventy degrees. We also return the average rate of the four raptor rates to see if there is any correlation between all four of the species.

Grammar Grove in 2004

```

--Rate of raptor sightings compared to temperature at Grammar Grove in 2004
SELECT gg.date, gw.max_temp, (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
(gg.noeha * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofnorthernharriersightings,
(gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofbald eaglesightings,
(gg.rsba * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
(((gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.noeha * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.rsba * 1.00) / NULLIF(gg.obs * 1.00, 0)) / 4) AS averagerate
--Return the date, average temperature, and calculate the rate of raptor sightings per hour as the count of bald eagles observed divided by the number of observation hours logged.
--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
FROM project.gg_count AS gg, project.gw_weather AS gw
--data is pulled from gg_count, which contains the date, count of raptors, and number of hours observed per day
--data is pulled from gw_weather, which contains the daily max temperature
WHERE gg.date = gw.date AND gw.max_temp < 70 AND (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND gg.date >= '2004-01-01' AND gg.date <= '2004-12-31'
--gg_count and gw_weather are joined by the date column
--filters out the days where the max temperature is above 70 degrees
--Ensures that zeros that are returned are not considered as a null value
--Ensures that data is only pulled from the year 2004 (specifying year reduces the amount of variables that could be influencing rate of raptor migration)
Order by max_temp desc;

```

Grammar Grove in 2024

```

--Rate of raptor sightings compared to temperature at Grammar Grove in 2024
SELECT gg.date, gw.max_temp, (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
(gg.noeha * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofnorthernharriersightings,
(gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofbald eaglesightings,
(gg.rsba * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
(((gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.noeha * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.rsba * 1.00) / NULLIF(gg.obs * 1.00, 0)) / 4) AS averagerate
--Return the date, average temperature, and calculate the rate of raptor sightings per hour as the count of bald eagles observed divided by the number of observation hours logged.
--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
FROM project.gg_count AS gg, project.gw_weather AS gw
--data is pulled from gg_count, which contains the date, count of raptors, and number of hours observed per day
--data is pulled from gw_weather, which contains the daily max temperature
WHERE gg.date = gw.date AND gw.max_temp < 70 AND (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND gg.date >= '2024-01-01' AND gg.date <= '2024-12-31'
--gg_count and gw_weather are joined by the date column
--filters out the days where the max temperature is above 70 degrees
--Ensures that zeros that are returned are not considered as a null value
--Ensures that data is only pulled from the year 2024 (specifying year reduces the amount of variables that could be influencing rate of raptor migration)
Order by max_temp desc;

```

Hitchcock in 2004

```

1--Rate of raptor sightings compared to temperature at Hitchcock in 2004
2 SELECT hc.date, hw.avg_temp, (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
3 (hc.noah * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.noah * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0)) / 4) AS averagerate
7--Return the date, average temperature, and calculate the rate of raptor sightings per hour as the count of bald eagles observed divided by the number of observation hours logged.
8--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.hitchcock_count AS hc, project.hc_weather AS hw
10--data is pulled from hitchcock_count, which contains the date, count of each raptor species, and number of hours observed per day
11--data is pulled from hc_weather, which contains the daily average temperature
12 WHERE hc.date = hw.date AND hw.avg_temp < 70 AND (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND hc.date >= '2004-01-01' AND hc.date <= '2004-12-31'
13--hitchcock_count and hc_weather are joined by by the date column
14--filters out the days where the average temperature is above 70 degrees
15--Ensures that zeros that are returned are not considered as a null value
16--Ensures that data is only pulled from the year 2004 (specifying year reduces the amount of variables that could be influencing rate of raptor migration
17 Order by avg_temp desc;
```

Hitchcock in 2024

```

1--Rate of raptor sightings compared to temperature at Hitchcock in 2024
2 SELECT hc.date, hw.avg_temp, (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
3 (hc.noah * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.noah * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0)) / 4) AS averagerate
7--Return the date, average temperature, and calculate the rate of raptor sightings per hour as the count of bald eagles observed divided by the number of observation hours logged.
8--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.hitchcock_count AS hc, project.hc_weather AS hw
10--data is pulled from hitchcock_count, which contains the date, count of each raptor species, and number of hours observed per day
11--data is pulled from hc_weather, which contains the daily average temperature
12 WHERE hc.date = hw.date AND hw.avg_temp < 70 AND (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND hc.date >= '2024-01-01' AND hc.date <= '2024-12-31'
13--hitchcock_count and hc_weather are joined by by the date column
14--filters out the days where the average temperature is above 70 degrees
15--Ensures that zeros that are returned are not considered as a null value
16--Ensures that data is only pulled from the year 2024 (specifying year reduces the amount of variables that could be influencing rate of raptor migration
17 Order by avg_temp desc;
```

After pulling data for temperature patterns, we used very similar queries to find out if there are any patterns related to precipitation level. We pulled the rate of observation of each of the four primary raptor species compared to the days of the highest amount of precipitation, along with the average rate of the raptors. In order to run the query, we also had to specify within the SQL code that any days of zero precipitation were not null values, but rather days where weather was observed and there was actually zero precipitation.

Grammar Grove in 2004

```

1--Rate of raptor sightings compared to precipitation level at Grammar Grove in 2004
2 SELECT gg.date, gw.precip, (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
3 (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0)) / 4) AS averagerate
7--Return the date, amount of precipitation, and calculate the rate of raptor sightings per hour as the count of raptors observed divided by the number of observation hours logged.
8--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.gg_count AS gg, project.gg_weather AS gw
10--data is pulled from gg_count, which contains the date, count of each raptor species, and number of hours observed per day
11--data is pulled from gg_weather, which contains the daily amount of precipitation
12 WHERE gg.date = gw.date AND gw.precip LIKE '%.%'
13 AND (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL
14 AND (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND gg.date >= '2004-01-01' AND gg.date <= '2004-12-31'
15--gg_count and gg_weather are joined by by the date column
16--precipitation levels above 0 (all precip levels above zero contain a decimal point)
17--Ensures that data is only pulled from the year 2004
18 ORDER BY precip;
```

Grammar Grove in 2024

```

1--Rate of raptor sightings compared to precipitation level at Grammar Grove in 2024
2 SELECT gg.date, gw.precip, (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofperegrinfalcon sightings,
3 (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) + (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0)) / 4) AS averagerate
7--Return the date, amount of precipitation, and calculate the rate of raptor sightings per hour as the count of raptors observed divided by the number of observation hours logged.
8--NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.gg_count AS gg, project.gg_weather AS gw
10--data is pulled from gg_count, which contains the date, count of each raptor species, and number of hours observed per day
11--data is pulled from gg_weather, which contains the daily amount of precipitation
12 WHERE gg.date = gw.date AND gw.precip LIKE '%.%'
13 AND (gg.pefa * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND (gg.noah * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND (gg.baea * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL
14 AND (gg.rsha * 1.00) / NULLIF(gg.obs * 1.00, 0) IS NOT NULL AND gg.date >= '2024-01-01' AND gg.date <= '2024-12-31'
15--gg_count and gg_weather are joined by by the date column
16--precipitation levels above 0 (all precip levels above zero contain a decimal point)
17--Ensures that data is only pulled from the year 2024
18 ORDER BY precip;
```

Hitchcock in 2004

```

1--Rate of raptor sightings compared to precipitation level in a given year at Hitchcock in 2004
2 SELECT hc.date, hw.precip, (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofperegrinalconsightings,
3 (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0)) / 4) AS averagerate
7 --Return the date, amount of precipitation, and calculate the rate of raptor sightings per hour as the count of raptors observed divided by the number of observation hours logged.
8 --NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.hitchcock_count AS hc, project.hw_weather AS hw
10 --data is pulled from hitchcock_count, which contains the date, count of each raptor species, and number of hours observed per day
11 --data is pulled from hw_weather, which contains the daily amount of precipitation
12 WHERE hc.date = hw.date AND hw.precip LIKE '1.1'
13 AND (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL
14 AND (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND hc.date >= '2004-01-01' AND hc.date <= '2004-12-31'
15 --hitchcock_count and hw_weather are joined by the date column
16 --precipitation levels above 0 (all precip levels above zero contain a decimal point)
17 --Ensures that data is only pulled from the year 2004 (specifying year reduces the amount of variables that could be influencing rate of raptor migration)
18 ORDER BY precip;

```

Hitchcock in 2024

```

1--Rate of raptor sightings compared to precipitation level in a given year at Hitchcock in 2024
2 SELECT hc.date, hw.precip, (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofperegrinalconsightings,
3 (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofnorthernharriersightings,
4 (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofbald eaglesightings,
5 (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) AS rateofredshoulderedhawksightings,
6 (((hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) + (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0)) / 4) AS averagerate
7 --Return the date, amount of precipitation, and calculate the rate of raptor sightings per hour as the count of raptors observed divided by the number of observation hours logged.
8 --NULLIF function is used to ensure that division by zero doesn't happen if the observation hours is zero on a given day
9 FROM project.hitchcock_count AS hc, project.hw_weather AS hw
10 --data is pulled from hitchcock_count, which contains the date, count of each raptor species, and number of hours observed per day
11 --data is pulled from hw_weather, which contains the daily amount of precipitation
12 WHERE hc.date = hw.date AND hw.precip LIKE '1.1'
13 AND (hc.pefa * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND (hc.noha * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND (hc.baea * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL
14 AND (hc.rsha * 1.00) / NULLIF(hc.obs * 1.00, 0) IS NOT NULL AND hc.date >= '2024-01-01' AND hc.date <= '2024-12-31'
15 --hitchcock_count and hw_weather are joined by the date column
16 --precipitation levels above 0 (all precip levels above zero contain a decimal point)
17 --Ensures that data is only pulled from the year 2024 (specifying year reduces the amount of variables that could be influencing rate of raptor migration)
18 ORDER BY precip;

```

One additional query we conducted found specifically the hourly rate of bird sightings by species and year. This informed us of expected sightings of a given species per hour derived from yearly total counts, divided by yearly total hours, to account for outlier days in either bird sightings or hours.

```

SELECT
    EXTRACT(YEAR FROM date) AS yr,

    -- Observed hours in the year
    SUM(gg_obs) AS total_gg_hours,
    SUM(h_obs) AS total_h_hours,

    -- Observed birds in the year
    SUM(gg_count) AS total_gg_birds,
    SUM(h_count) AS total_h_birds,

    -- Rate of birds per hour in the year
    SUM(gg_count)::numeric / NULLIF(SUM(gg_obs), 0) AS gg_rate,
    SUM(h_count)::numeric / NULLIF(SUM(h_obs), 0) AS h_rate

FROM project.noha
GROUP BY EXTRACT(YEAR FROM date)
ORDER BY yr;

```

Results

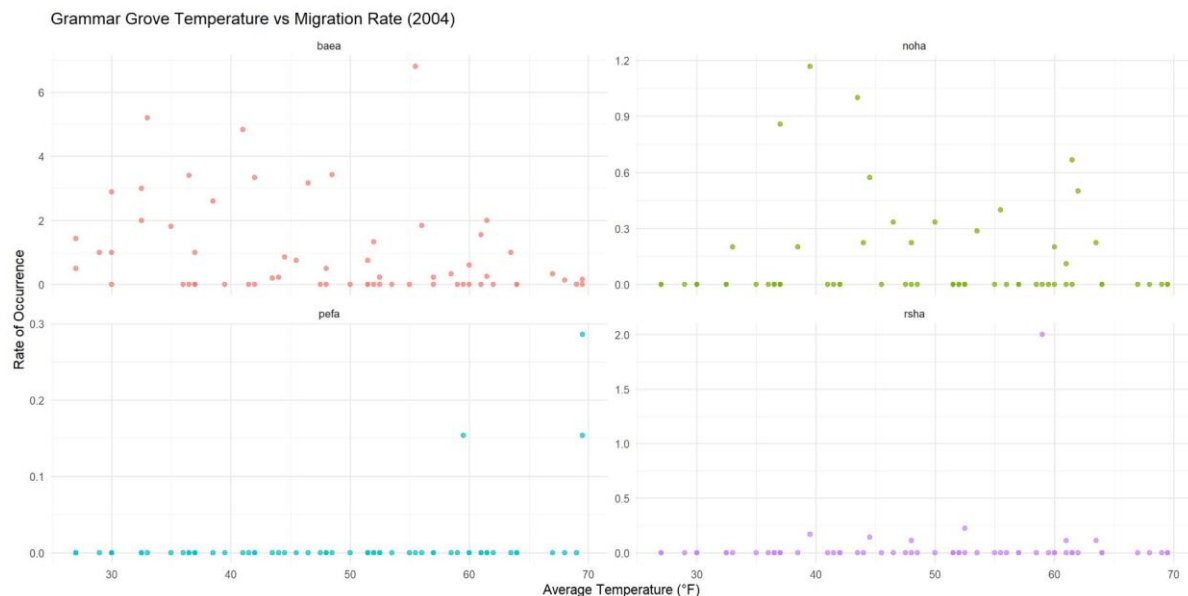
We were able to learn a lot from the results of both our spatial and non-spatial queries. First, both Hitchcock and Grammar Grove are located a considerable distance from any bird conservation area. The closest conservation area to Hitchcock is the Iowa River Corridor, thirty-five kilometers away, and the closest area to Grammar Grove is Loess Hills, twenty-six kilometers away. Since conservation areas are a significant distance away from watch sites, it makes it difficult to draw any definitive conclusions from this query. Based on the information

that we have, it is very unlikely that any conservation practices twenty-six kilometers away from the raptor observation sites have any measurable impact.

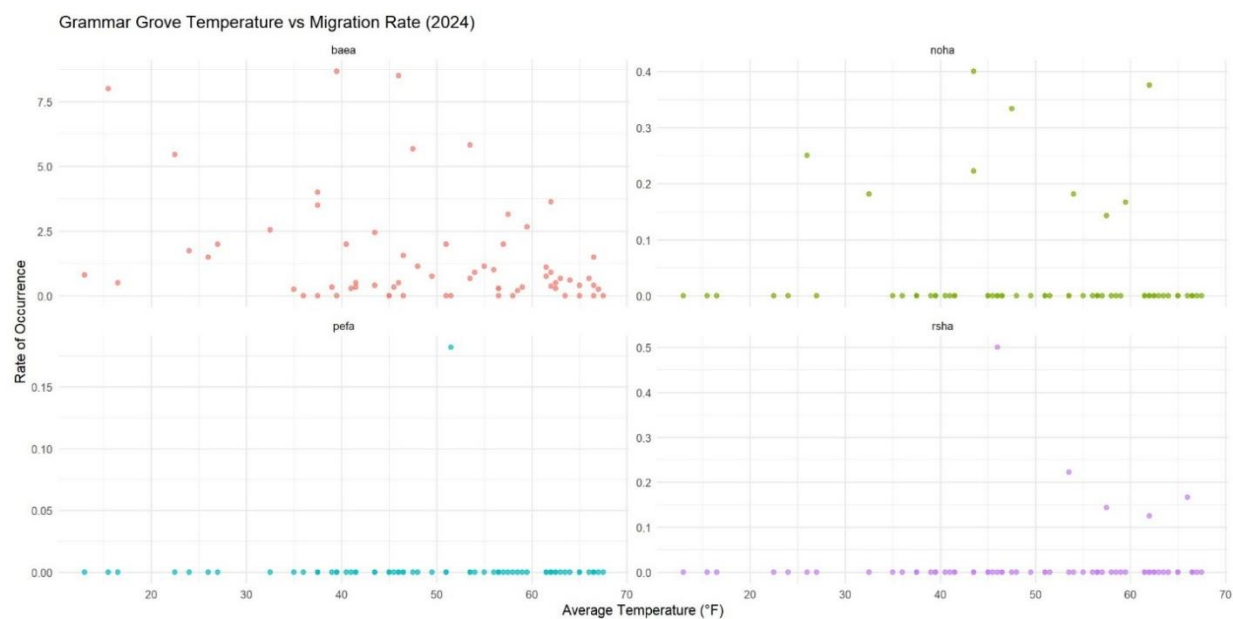
	bird_conservation_area	distance_km_grammar_grove		bird_conservation_area	distance_km_hitchcock
1	Iowa River Corridor	35.45288641435178	1	Loess Hills	25.578003101601052
2	Chichaqua-Neal Smith	46.98194600104233	2	Raccoon River	100.79309016788666
3	Boone Forks	65.28145951093978	3	Broken Kettle Grasslands	140.625142245963
4	Lower Morse Lake	80.16167745396663	4	Kellerton	156.19105409477993
5	Wapsi River	85.95482865690678	5	Waterman Prairie	158.87804071137688
6	Union Hills	92.61376013728591	6	Sand Creek	159.85463665026802
7	Stephens Forest-1,000 Acre Unit	107.05658324233345	7	Boone Forks	175.97184806598085
8	Eagle Lake Wetlands	117.35861239497751	8	Stephens Forest	184.1794338582284
9	Stephens Forest	123.87903343289508	9	Dewey's Pasture	198.9141380071986
10	Raccoon River	124.59213013675313	10	Chichaqua-Neal Smith	203.53912709966136

As for the weather comparison queries, these had slightly more success. In order to properly visualize the returned data, we used R to create plots of the Red Shouldered hawk, Northern Harrier hawk, Peregrine falcon, and Bald eagle sightings in relation to the date and temperature. Each point on each respective plot represents one day's worth of observations of a given species, and points at zero on the Y axis represent days of no raptor sightings. Both Hawk Watch locations over the 20-year period show a moderate negative correlation between temperature and Bald eagle sightings. This means that, as temperature increases, Bald eagle sightings tend to decrease. The Northern harrier has an extremely weak negative correlation over both Hawk Watches, and the Peregrin falcon and Red Shouldered hawk have no observable relationship. While the number of raw raptor sightings tended to increase over time, the relationships between raptor species and temperature remained relatively the same.

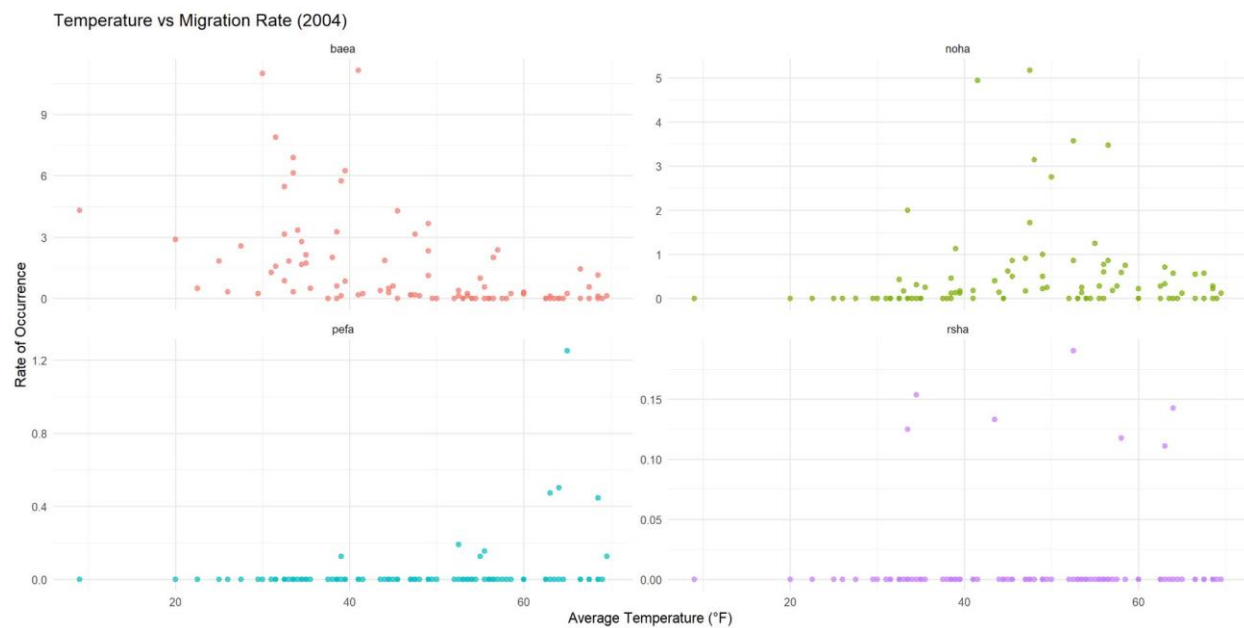
Grammar Grove 2004



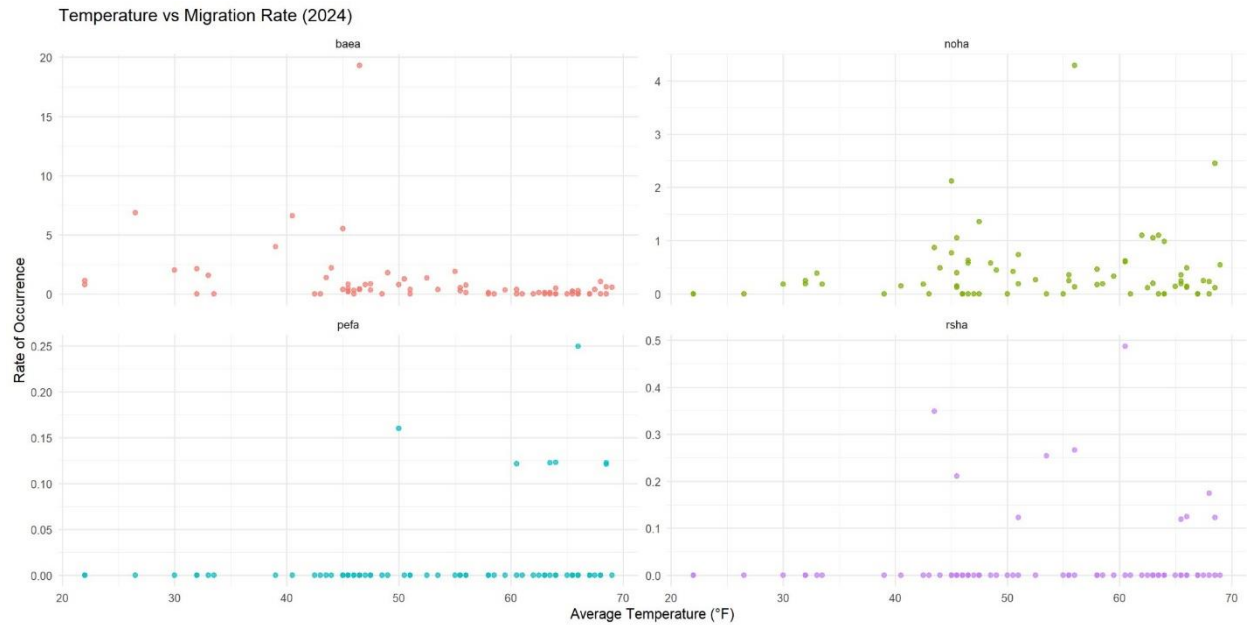
Grammar Grove 2024



Hitchcock 2004

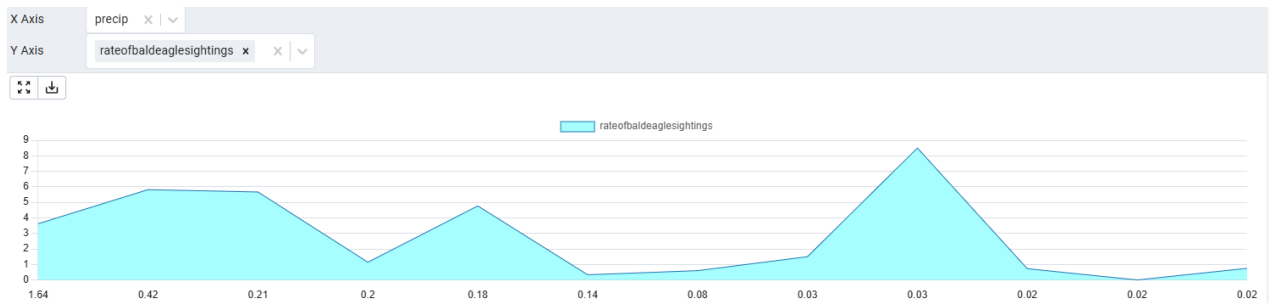


Hitchcock 2024

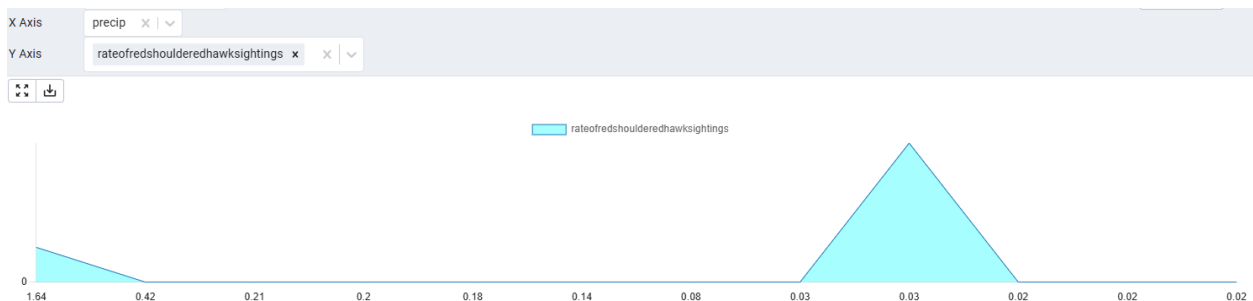


While we were able to see negative relationships between raptor sightings and temperature, these patterns were non-existent between raptor sightings and inches of precipitation. The rate of sightings varied greatly over different levels of precipitation. Examples are provided below for each raptor species at Grammar Grove Hawk Watch. As can be observed, there are many peaks and valleys within each species' observations when they are ordered by precipitation. Similar levels of stochasticity are observed at Hitchcock as well.

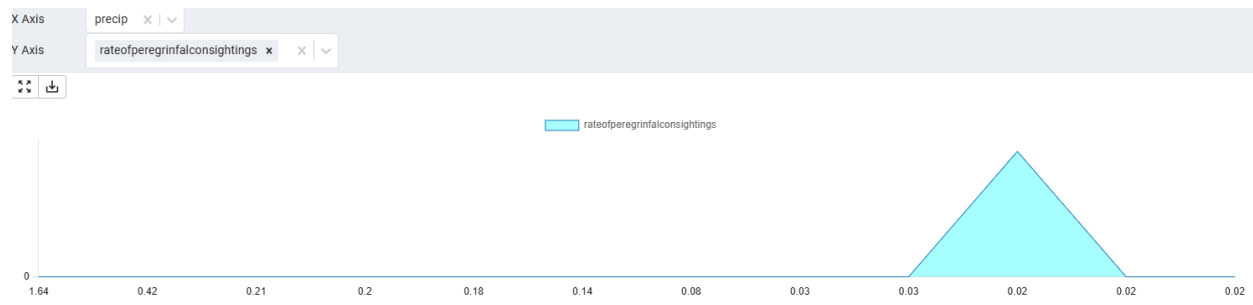
Bald Eagle sightings compared to precipitation (inches)



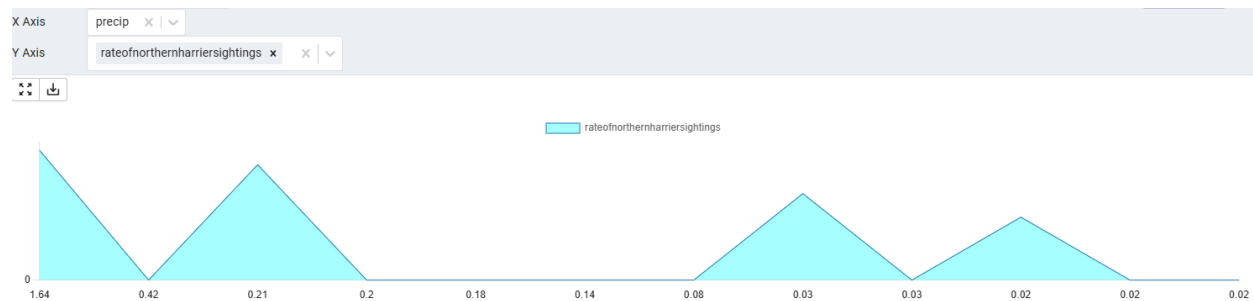
Red Shouldered hawk sightings compared to precipitation (inches)



Peregrin falcon sightings compared to precipitation (inches)



Northern harrier sightings compared to precipitation (inches)



Conclusion

For this project, we decided to analyze raptor sighting data in Iowa at two locations, Hitchcock overlooking migrating birds along the Missouri River and Grammar Grove viewing birds along the Iowa River. We were looking for a connection between the frequency of raptor sightings in relation to weather and location. Data included temperature, precipitation, and rate of occurrence between both locations. The query we ran, to see the relationship between raptor count and average daily temperature, included four raptor species at both locations in 2004 verses 2024. From this query, we found that there was not a strong correlation between temperature and rate of occurrence for most of the bird species over time. For the Bald Eagle, we found a slightly negative correlation between temperature and sightings suggesting that they prefer generally colder temperatures. For the Northern Harrier we also found a negative correlation, but since it is a very weak correlation, we cannot say with certainty that they prefer colder temperatures. For the other two species, we observed there was no correlation between temperature and observance. Next for precipitation, we also did not find any correlation between frequency of observations and amount of precipitation.

Throughout this project, many things went well. The data was easy to clean, and our database structure was simple because there were only two-point locations. By narrowing our scope to only analyze the patterns of the four raptor species of special interest in Iowa, it allowed us to focus more on the structure, then get overwhelmed by the sheer quantity of data. There are multiple limitations in our data. Specifically, the raptor data itself is collected by human viewers who cannot man observation locations 24/7, and volunteers/workers can easily misidentify raptors that look similar. Because these two Hawk Watch locations were only in Iowa, we don't have a frame of reference for what these raptor migration patterns look like up or down river.

There are many factors that we could include, if given more time, improving the quality of our data. We included bodies of water and historical vegetation to our project to show factors that could contribute to the data shown but were unable to execute more queries due to the complexity of the project and lack of time. Other factors that could be included in the future that may influence frequency include urban areas, policy, pollution, and prey population. We could also increase the scope to include the entire United States instead of just looking at Iowa.

Although there were no strong correlations with the data points we were looking at, we were still able to gain some insight into raptor migratory patterns along the Missouri River and Iowa River. One beneficial factor about the project is that we were most likely the only people to gather data on raptor flight in comparison to other factors along the Northern part of the Missouri River. Ideally, if this database were public and fully formed and operating, it would provide influential insights on factors contributing to raptor counts. This could inform policy for raptor conservation areas in the future by following potential changes in migration patterns. Having an easily accessible raptor migration database also allows for other ecological studies. Because raptors are an apex predator, changes in population, migration pattern, or behavior can relate heavily to ecosystem health. This database may allow for comparison and information that can offer insight for ecological research. This project has established a framework for a raptor database, and in the future, it could be applied in a broader context to include a wider range of data leading to correlative and conclusive information on raptor migration.

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